The Chain Rule

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic chain rule applications

Differentiate each function using the chain rule:

a $f(x) = (4x - 1)^5$

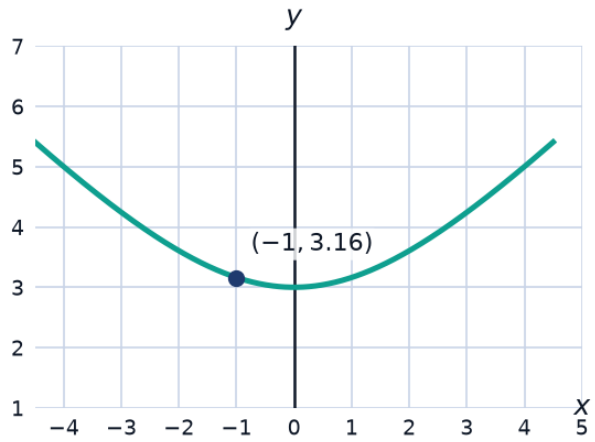
b $f(x) = \sin(3x^2)$

c $f(x) = e^{5x}$

d $f(x) = \ln(2x + 7)$

2 [Foundational] • Target: Basic chain rule applications

Differentiate $f(x) = \sqrt{x^2 + 9}$.



3 [Foundational] • Target: The chain rule with trig and exponential functions

Differentiate $f(x) = \cos^3(x)$ (i.e. $[\cos x]^3$).

4 [Foundational] • Target: Nested (multiple) chain rule

Differentiate $f(x) = \sin\left(\sqrt{x^2 + 1}\right)$, applying the chain rule twice.

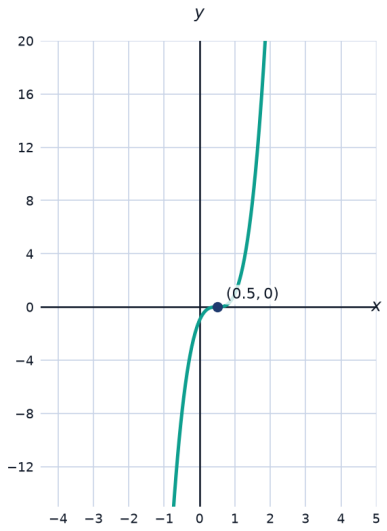
5 [Foundational] • Target: Chain rule with tables of values

Given $f(2) = 5$, $f'(2) = 3$, $g(2) = 7$, $g'(2) = -1$, find $\frac{d}{dx}[f(g(x))]$ at $x = 2$ if $g(2) = 7$ and $f'(7) = 4$ (note the composition uses f' evaluated at $g(2) = 7$, not at 2).

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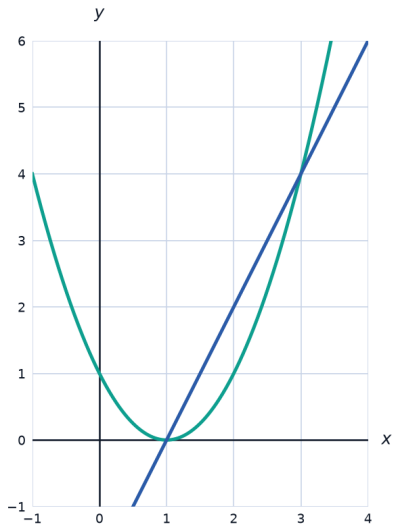
6 [Foundational] • Target: Finding tangent lines using the chain rule

Find the equation of the tangent line to $f(x) = (2x - 1)^3$ at $x = 1$.



7 [Foundational] • Target: Visualizing a composite function and its derivative

The graph shows $f(x) = (x - 1)^2$ (an inner-outer composite of $u = x - 1$ and u^2), with its tangent line at $x = 2$.



a Use the chain rule to find $f'(x)$.

b Confirm $f'(2)$ matches the slope of the tangent line shown.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: The chain rule in applied rate problems

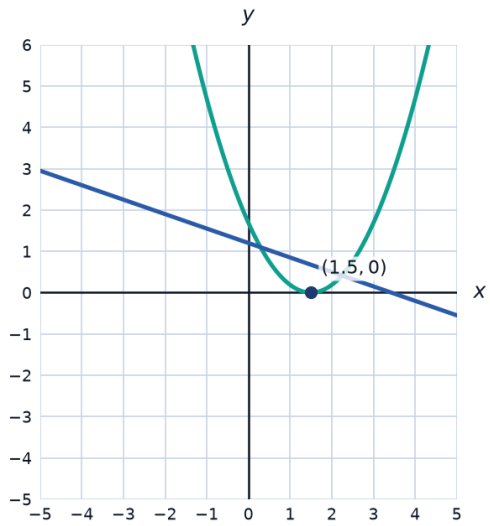
The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the radius is growing according to $r(t) = 2t + 1$ (cm, seconds). Use the chain rule to find $\frac{dV}{dt}$ at $t = 2$.

9 [Intermediate] • Target: Chain rule with a table of composed values

Given $h(x) = f(g(x))$ where $g(1) = 3$, $g'(1) = 2$, $f(3) = 7$, $f'(3) = -4$, find $h'(1)$.

10 [Intermediate] • Target: Chain rule with a table of composed values

Differentiate $f(x) = \sqrt[3]{x^2 + 1}$ using the chain rule.



11 [Intermediate] • Target: Basic chain rule applications

[Unit 8 · Applications of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

12 [Intermediate] • Target: The chain rule with trig and exponential functions

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

13 [Intermediate] • Target: Nested (multiple) chain rule

[Unit 6 · Integration and Accumulation of Change · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 [Intermediate] • Target: Chain rule combined with product/quotient rules

[Unit 7 · Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

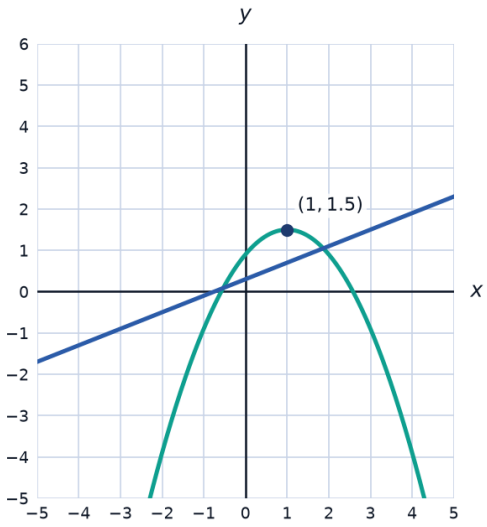
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15 **[Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

16 **[Challenging] • Target: Chain rule with inverse trig and logarithms • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \le 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.



17 **[Challenging] • Target: Finding tangent lines using the chain rule • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 1 · FoundationsFor the parametric curve $x(t) = \cos(t)$ and $y(t) = \sin(2t)$, find all values of $t \in [0, \pi]$ where the curve has a vertical tangent line.

18 **[Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1 + \tan x} - \sqrt{1 + \sin x}}{x^3}$.

19 **[Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization**

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

20 **[Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic chain rule applications

Use the chain rule to differentiate $f(x) = (x - 1)^2$, then find $f'(1)$.

a. 0

b. -2

c. 3

d. 5

22 [Foundational] • Target: The chain rule with trig and exponential functions

Which is the correct derivative of $F(x) = \sin(5x + 2)$?

a. $5\cos(5x + 2)$

b. $-5\sin(5x + 2)$

c. $5\cos(x)$

d. $\cos(5x + 2)$

23 [Foundational] • Target: Nested (multiple) chain rule

The side of a square is increasing at 2 cm/s. Find the rate of change of its area when the side length is 5 cm.

a. 18

b. 10

c. 20

d. 22

24 [Foundational] • Target: Chain rule combined with product/quotient rules

The radius of a circle increases at a rate of 3 cm/s. Find the rate of change of the area $\left(\frac{dA}{dt}\right)$ when $r = 17$ cm.

a. 34π

b. 102π

c. 102

d. 51π

25 [Foundational] • Target: Chain rule with tables of values

The region between $y = 2$ and $y = x^2$ on $[0, 1]$ is rotated about the x-axis. Find the volume using the washer method.

a. $\frac{19}{5}$

b. $\frac{19}{10}\pi$

c. $\frac{19}{5}\pi$

d. $\frac{38}{5}\pi$

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26 [Foundational] • Target: Chain rule with inverse trig and logarithms

Find the x-coordinate of the inflection point of $f(x) = x^3 - 5x^2$.

a. $\frac{5}{3}$

b. $\frac{8}{3}$

c. $\frac{2}{3}$

d. $-\frac{5}{3}$

27 [Foundational] • Target: Finding tangent lines using the chain rule

The region under $y = 1\sqrt{x}$ on $[0, 4]$ is rotated about the x-axis. Find the volume of the resulting solid.

a. 16π

b. 8π

c. 4π

d. 8

28 [Foundational] • Target: Visualizing a composite function and its derivative

If $f(x) = 1x^2 + 3x$, is f increasing or decreasing at $x = 0$?

a. decreasing

b. cannot be determined

c. constant

d. increasing

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The chain rule in applied rate problems

Evaluate the definite integral of the constant function: $\int_{8}^{11} 21\,dx$.

a. 21

b. 84

c. 42

d. 63

30 [Intermediate] • Target: Chain rule with a table of composed values

Find the area under the curve $f(x) = 1x^2 + 1$ on $[1, 2]$.

a. $\frac{16}{3}$

b. $-\frac{10}{3}$

c. $\frac{4}{3}$

d. $\frac{10}{3}$

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31 [Intermediate] • Target: Basic chain rule applications

Find the critical number of $f(x) = 5x^2 + 0x$.

a. 0

b. -1

c. 1

d. 2

32 [Intermediate] • Target: The chain rule with trig and exponential functions

A function is continuous at $x = a$ if:

a. $f(a)$ exists

b. $\lim_{x \to a} f(x)$ exists

c. Both exist and are equal

d. None of the above

33 [Intermediate] • Target: Nested (multiple) chain rule

Find the area bounded by $f(x) = 2x$, the x-axis, and the line $x = 6$, for $x \geq 0$.

a. 34

b. 38

c. 12

d. 36

34 [Intermediate] • Target: Chain rule combined with product/quotient rules

The diagram shows the graph of the function $y = f(x)$. The equation of the function is given by $f(x) = ax^2 + bx + c$. Given that the vertex of the parabola is at $(-1, 8)$ and the graph passes through the point $(1, 0)$, find the values of a and b.

a. a = -2, b = -4

b. a = -2, b = 4

c. a = 2, b = 4

d. a = 2, b = -4

35 [Intermediate] • Target: Chain rule with tables of values

Consider the region R bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π

d. $\frac{\pi}{4}$

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36 [Intermediate] • Target: Chain rule with inverse trig and logarithms

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y \sqrt{1 + (y')^2}\, dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

b. π

c. 2π

d. 4π

37 [Intermediate] • Target: Finding tangent lines using the chain rule

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s?

a. 8 m²/s

b. 16 m²/s

c. 32 m²/s

d. 4 m²/s

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing a composite function and its derivative • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The integral $\int_0^1 e^{-x^2} dx$ (error function) — the exact value cannot be expressed in elementary functions. Using the series, approximate as $\sum \frac{(-1)^n}{n!\,(2n+1)}$. First two terms:

a. $1 - 1/6 \approx 0.833$

b. $1 - 1/3 \approx 0.667$

c. $1 + 1/3 \approx 1.333$

d. 0

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39 [Challenging] • Target: The chain rule in applied rate problems • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$

a. $3(x^2-2)+2x(3x+1)$

b. $3x^2-6+6x^2+2x$

c. $9x^2+2x-6$

d. all are equivalent

40 [Challenging] • Target: Chain rule with a table of composed values • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

If $f(x) = \frac{4}{x}$ (a self-inverse function), find the positive value of x for which $f(x) = x$ (a fixed point).

a. 2

b. 1

c. 3

d. 4

41 [Challenging] • Target: Basic chain rule applications • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

A function has exactly one local min and no local max on (a, b). Its f' must change from:

a. $-$ to $+$

b. $+$ to $-$

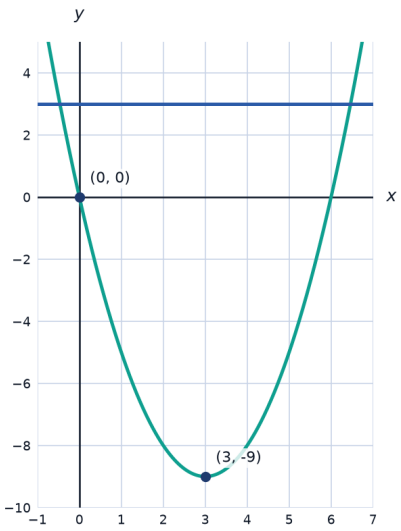
c. $+$ to $+$

d. 0 to $+$

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42 [Challenging] • Target: The chain rule with trig and exponential functions • Cross-Lesson: 7.01 Differential Equations and Slope Fields

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.



a. $a = 1, c = 0$

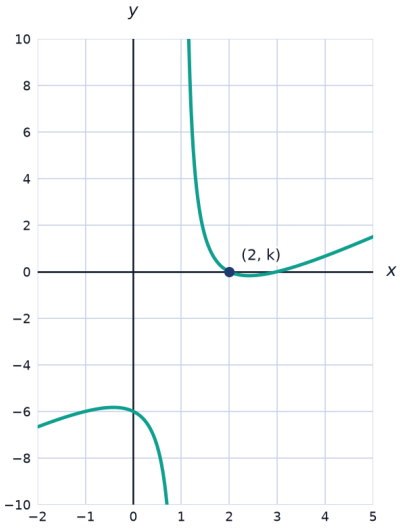
b. $a = 2, c = 0$

c. $a = 1, c = 3$

d. $a = 2, c = 3$

43 [Challenging] • Target: Nested (multiple) chain rule • Cross-Lesson: 8.01 Area Between Curves

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.



a. 4

b. 5

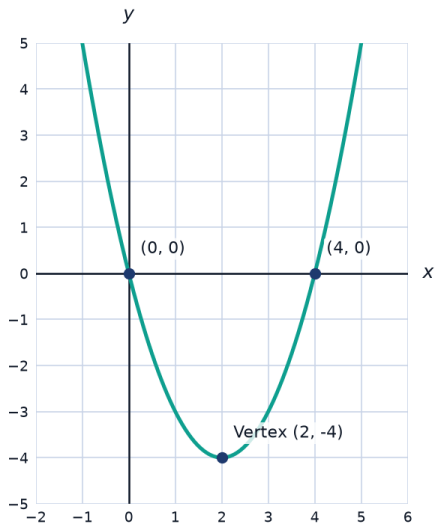
c. 6

d. 7

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44 **[Challenging]** • **Target: Chain rule combined with product/quotient rules** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.



a. -1

b. 1

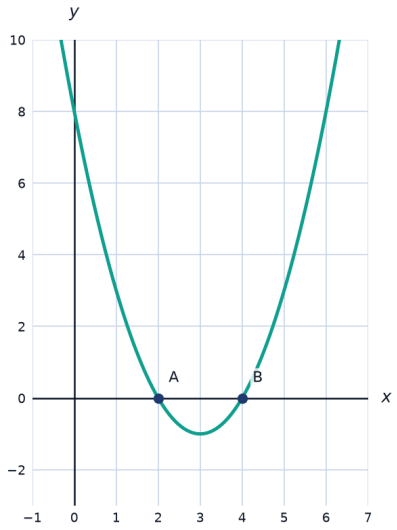
c. -2

d. 2

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45 [Challenging] • Target: Chain rule with tables of values • Cross-Lesson: 1.01 Limits Graphically and Numerically

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?



a. (2, 0)

b. (4, 0)

c. (0, 8)

d. (1, 3)

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